

1
2 Personalized AI assistants could either promote or hinder the
3 emergence of collective intelligence

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9 **Abstract**

10 Advances in artificial intelligence (AI) have enabled the wide use of personalized
11 assistance for cognitive tasks, which enhanced individual performance on various
12 fields. However, the impact of AI assistance on collective intelligence remains poorly
13 understood. AI assistance may undermine the emergence of collective intelligence by
14 reducing the individual diversity or introducing structured biases into a population.
15 Here, we develop a human-AI collective prediction model mediated by social learning
16 based on incentive structure by extending the existing works on collective intelligence.
17 Our model uncovers that emergence of collective intelligence is depend on how human
18 and AI interact. When AI works as if it is omniscient and gives assistance regardless
19 of users' interests, collective intelligence will be promoted. However, when AI works
20 as a chat-bot and gives assistance only based on users' interests, collective intelligence
21 could be hindered due to diversity loss and overconfidence on AI. We showed that,
22 to promote collective intelligence, penalizing or incentivizing AI usage is not reliable.
23 Instead, elaborating more diversity-aware incentives could be the key to promote and
24 maintain the collective intelligence.

Introduction

Collective intelligence (CI) describes a phenomenon in which a group’s ability to solve problems, make decisions, or adapt to changing environments exceeds each member’s ability [1, 2, 3, 4, 5]. This phenomenon, also referred to as wisdom of crowds, has thus served as a central mechanism underlying the success of diverse organizations, ranging from animal groups [6, 7, 8] and human societies [2, 9] to machine-learning systems [10, 11]. Understanding the conditions that promote and maintain CI therefore remains a fundamental question across social sciences [12], evolutionary biology [13], and computer science [14]. Addressing this question has become particularly timely as artificial intelligence (AI) has begun to reshape how individuals acquire information, make decisions, and contribute to collective outcomes [15].

The recent emergence of generative AIs and large language models (LLMs) has changed the way individuals engage in various cognitive tasks, ranging from simple everyday tasks to more complex tasks, including scientific discovery [16] and creative writing [17]. In particular, while the use of generative AI models, such as ChatGPT, has been rapidly increasing [18], the potential long-term impact of personal AI assistants on human society, especially the CI, remains uncertain. Some argue that AI can promote CI by enhancing individual expertise [19], and opinion aggregation [20]. Conversely, empirical evidence indicate that the use of AIs can limit the diversity of opinions [17, 21, 22], a key requirement for the emergence of CI [23, 24, 25].

Our ability to predict the long-term impact of personal AI assistants on human CI is further hampered by the apparent discrepancy in the incentives that shape AI uses. On one hand, many industrial sectors are pushing towards increased AI usages, where the number of “tokens” used is often perceived as a proxy for productivity [26, 27]. On the other hand, many academic institutions appear to discourage, and even penalize, AI uses, reflecting concerns about plagiarism, misinformation, and limited independence [28, 29]. Such discrepancy raises a question about whether directly incentivizing or penalizing AI usage will benefit or harm human CI. These observations further highlight the importance of identifying a robust incentive structure that can promote CI in the presence of personal AI assistants. Such incentive structure should seek to retain the benefits of AIs and prevent potential risk, such as preserving the diversity of individual opinions.

To address these questions, we extend previous approaches for modeling CI, where each individual makes independent predictions for the outcome from partial observations of underlying causal factors [12, 13, 30]. In particular, we allow each individual to independently consult with personal AI assistants and revise their predictions accordingly. The resulting model allows us to investigate the robustness of human CIs under different assumptions about the AI model’s ability to make predictions as well as incentive structures that are associated with the prediction task. In doing so, we assess the impact of directly incentivizing or penalizing AI uses on CI and further suggest a balanced incentive structure that can maintain CI in the presence of AI.

66 Main

67 Model

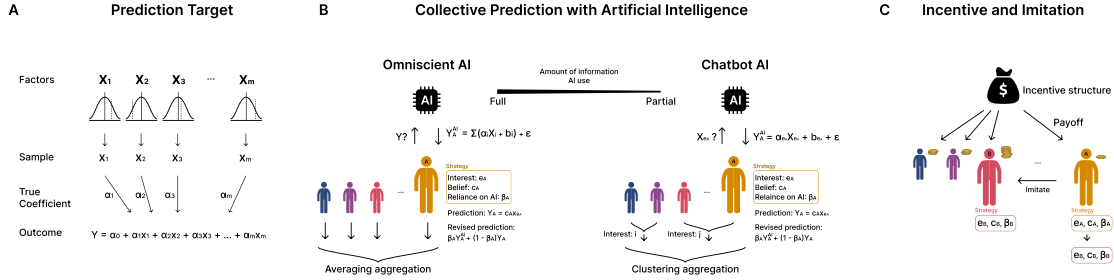


Figure 1: **A schematic diagram of the model for individual and collective prediction in the presence of personal AI assistants.** (A) The prediction target is modeled as a linear combination of causal factors. (B) Each player is characterized by three strategies: interest, belief, and reliance on AI. Players make predictions based on their interests and beliefs and subsequently revise them with AI assistance. We consider two types of AI, Chatbot AI and Omniscient AI, which differ in the amount of information available for generating predictions. (C) Players receive payoffs according to the incentive structure, and the population evolves through imitation of peers with higher payoffs.

68 We begin by extending previously established models of CI, in which a group of
 69 players participates in a prediction task (Figure 1): each player makes an individual
 70 prediction based on partial observations of the underlying causal factors, and the
 71 group forms a collective prediction by aggregating these individual predictions [30,
 72 12, 13]. Specifically, the prediction target (Y) is modeled as a linear combination of
 73 M causal factors x_i , with fixed coefficients α_i : $Y = \sum_{i=1}^M \alpha_i x_i + \alpha_0$, where each factor
 74 is drawn from independent normal distribution (Figure 1A). Then, each individual
 75 A observes a single factor of their interest x_{e_A} and makes a prediction based on
 76 their own belief c_A : $Y_A = c_A x_{e_A}$ (Figure 1B). We assume each individual can then
 77 independently consult their AI assistants and revise their prediction based on their
 78 reliance on AI β_A : $Y_A^{re} = \beta_A Y_A^{AI} + (1 - \beta_A) Y_A$ (Figure 1B). The revised predictions are
 79 then aggregated using two alternative aggregation rules, averaging and clustering, to
 80 form collective prediction.

81 To model AI, we assume that AI has already been trained on similar tasks and is
 82 able to make predictions based on partial or full observations of causal factors (Figure
 83 1B). Every time a player consults AI, it will make a new prediction Y_A^{AI} based on
 84 its internal parameters, which can include factor-specific bias (b_i) and stochastic
 85 error ϵ . Here, we consider two extreme types of AI response, which differ in the
 86 amount of information AI uses when making predictions. In one extreme, the AI
 87 can observe all causal factors and use all information to make predictions, regardless

88 of the user’s interest: $Y_A^{AI} = \sum_{i=0}^M (\alpha_i x_i + b_i) + \epsilon$. In the other extreme, AI works
 89 like a chatbot, where the AI predicts the outcome based only on the user’s interest:
 90 $Y_A^{AI} = \alpha_{e_A} x_{e_A} + b_{e_A} + \epsilon$. Hereafter, we refer to these two types as ‘Omniscient AI’
 91 and ‘Chatbot AI’, respectively.

92 *[JS: We assume that appropriate aggregation rule for collective prediction de-*
 93 *pends on the type of AI. With Omniscient AI, each revised prediction contains infor-*
 94 *mation of all causal factors and therefore represents an estimate of the full prediction*
 95 *target. With Chatbot AI, each revised prediction remains an estimate of the contri-*
 96 *bution of a single factor to the target. Based on this distinction, we use averaging*
 97 *aggregation for Omniscient AI, in which revised individual predictions are simply*
 98 *averaged to form the collective prediction. For Chatbot AI, we use clustering aggre-*
 99 *gation, in which revised predictions are first averaged among individuals interested*
 100 *in the same factor, and then summed across factors.]*

101 Each generation consists of multiple round of predictions, and each player is re-
 102 warded a payoff according to the incentive structure (Figure 1C). For simplicity,
 103 we assume that a sufficiently large number of prediction rounds occurs within each
 104 generation, allowing us to use the expected payoff value rather than computing the
 105 payoffs for every round. We consider two incentive structures from previous work,
 106 “feedback” and “niche-expert”, which have been shown to promote near-perfect col-
 107 lective accuracy in the absence of AI [13]. The feedback structure gives higher payoff
 108 to individuals who can push the collective prediction toward the true target, regard-
 109 less of their individual predictive accuracy: *[JS: $\pi_A = Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}})$]*. In contrast, the
 110 niche-expert structure gives higher payoff to individuals who have both better pre-
 111 dictive accuracy and a minority interest: *[JS: $\pi_A = -\rho_{e_A}(Y_A^{\text{re}} - Y)^2$]*. *[JS: Here, ρ_{e_A}*
 112 *denotes the proportion of players interested in the same factor, thereby allowing the*
 113 *niche-expert structure to preserve interest diversity.]* We note collective intelligence
 114 fails to emerge under the combination of averaging aggregation with niche-expert
 115 structure in the absence of AI [13].

116 Through time, the group evolves by allowing each individual to imitate peers
 117 who received higher payoffs (Figure 1C). Specifically, after each generation, two in-
 118 dividuals are randomly sampled, and one imitates all three strategies (the factor of
 119 interest, factor belief, and reliance on AI) of the other with a probability determined
 120 by the difference in their expected payoffs *[JS: (see the Simulations section in Meth-*
 121 *ods)]*. Individual predictions are aggregated into a collective prediction $\hat{Y}$ based on
 122 the aggregation rules. Collective accuracy is defined as the reduction in prediction
 123 error achieved by the collective prediction relative to a baseline prediction α_0 : *[JS:*
 124 *$1 - \frac{\mathbb{E}[(Y - \hat{Y}^{\text{re}})^2]}{\mathbb{E}[(Y - \alpha_0)^2]}$]*. Hereafter, we say that CI emerges when the collective accuracy
 125 reaches a near perfect level.

126 This model now allows us to explore the impact of AI on CI depending on the
 127 mode of AI and the incentive structure. Here, we begin by comparing the emergence
 128 of CI with and without AI and subsequently investigate a how different incentive
 129 structures affect the emergence of CI in the presence of AI.

AI can either promote or prevent the emergence of CI

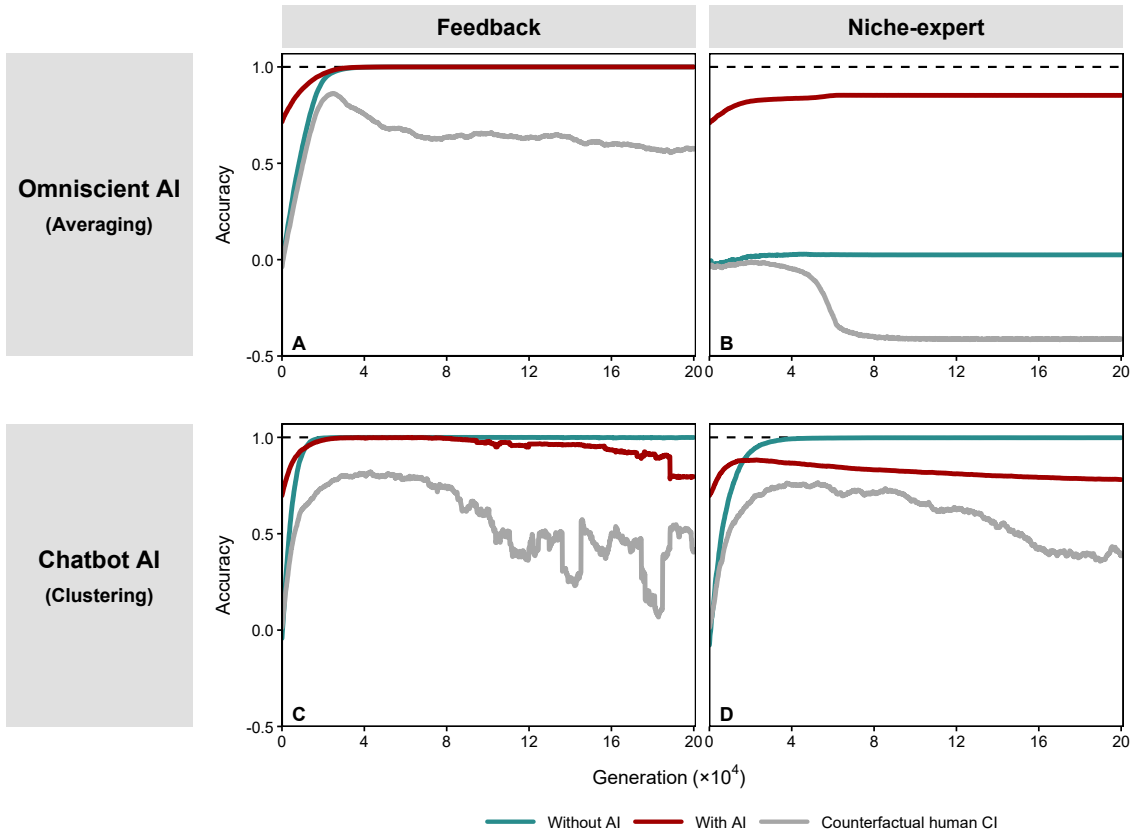


Figure 2: **Comparison of collective accuracy with and without AI.** Trajectories of collective accuracy when the group is assisted by the Omniscient AI (A,B) and Chatbot AI (C,D). [SWP: Maybe we should also name the averaging rule below AI...] Here, feedback (A,C) and niche-expert (B,D) incentive structures are considered. Green lines represent trajectories of collective accuracy without AI assistance. Red lines represent trajectories of collective accuracy with AI assistance. Gray lines represent trajectories of counterfactual collective accuracy, computed from players’ unrevised predictions, indicating the group’s predictive accuracy if AI assistance were unavailable at the given generation. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 100^2)$ for (A, B), $c_A \sim \mathcal{N}(0, 5^2)$ for (C, D), and $\beta_A \sim U(0, 1)$. AI parameters: $b_i = 0.36$ and $\tau = 0.3$.

To understand the impact of AI, we first compare collective dynamics with and without AI. First, consistent with [13], simple imitation rules and incentive structures allow collective intelligence to emerge in the absence of AI (Fig. 2A,C,D). One exception is the use of simple averaging with niche-expert incentive structures, under which the collective accuracy of the population remains near zero. [JS: This is

because enhancing individual accuracy causes individuals to predict close to the true coefficient of their interest, but simply averaging these predictions does not necessarily produce an accurate estimate of the full prediction target.][SWP: because...]
(Fig. 2B) [SWP: maybe need a supp figure].

We find that AI usage can have heterogeneous outcomes depending on its type. For example, the use of Omniscient AI can either preserve CI under the feedback incentive structure (Fig. 2A) or promote higher collective accuracy under the niche-expert incentive structure (Fig. 2B). These results are robust across different levels of AI accuracy and can be shown analytically by assuming a sufficiently large group of players (Fig. S1; see SI Evolutionary dynamics section).

In contrast, the use of Chatbot AI prevents the emergence of CI, thereby resulting in lower levels of collective accuracy (Fig. 2C, D). Under the feedback incentive structure, the use of Chatbot AI leads to a loss of diversity in the population's interest, making it impossible to account for effect of some causal factors (Fig. S1, S2). This causes an increase in collective variance and stochastic decrease in collective accuracy (Fig. 1, S1)[SWP: Supp]. We note that the loss of interest diversity may arise even in the absence of AI in finite populations (Fig. S3), but the use of Chatbot AI makes the group more vulnerable to such loss (Fig. S3, S4). Under the niche-expert incentive structure, the population becomes overly reliant on AI, causing the collective prediction to inherit the AI's bias (Fig. S2). This leads to an increase in collective bias and a decrease in collective accuracy (Fig. 2, S2) [SWP: Supp]. Interestingly, we find that the niche-expert structure can prevent the use of AI under specific assumptions about the relationship between the true coefficient and AI bias (Fig. S1). In this case, the population completely excludes AI assistance, allowing the CI to emerge eventually. However, this phenomenon can substantially delay the emergence of CI (Fig. S5) [SWP: Supp].

To investigate whether the group can retain its predictive ability when AI assistance suddenly becomes unavailable at any given time, we quantify counterfactual collective accuracy based on unrevised predictions. Across all scenarios, we find that the counterfactual collective accuracy is lower than the collective accuracy of the corresponding population.

In summary, an omniscient AI can promote CI, but AI operating with limited information may prevent the emergence CI. Moreover, reliance on AI can limit inherent ability of the population to make accurate predictions on their own, raising concerns about future AI usage. These findings further underscore the need for incentive structures that can mitigate the potential risks associated with AI assistance.

Directly incentivizing or penalizing AI use can additionally harm CI

Given the strong influence of AI on collective outcomes, a straightforward intervention is to directly incentivize or penalize the use of AI. [SWP: maybe a sentence about how this reflects real world scenarios.] To investigate whether such direct

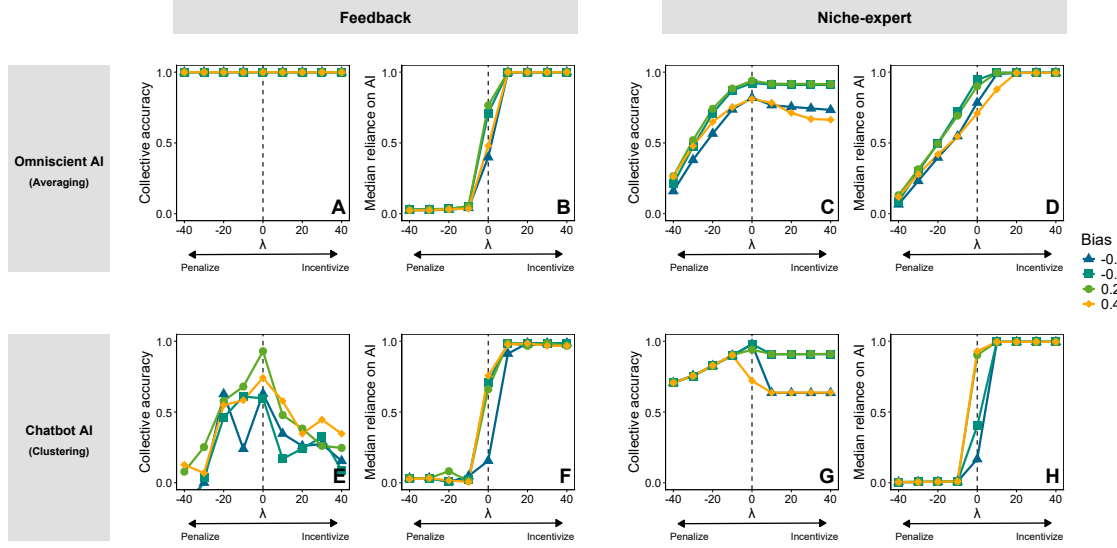


Figure 3: **Collective accuracy under the influence of direct intervention on AI use.** Collective accuracy and median reliance on AI under varying strengths of intervention (λ) across different assumptions about AI types (omniscient AI, A–D and chatbot AI, E–H) and incentive structures (Feedback, A,B,E,F, and Niche expert, C,D,G,H). Positive λ represents an incentive for AI use, whereas negative λ represents a penalty for AI use. In each panel, the vertical dashed line indicates the original incentive structure ($\lambda = 0$). Four AI bias levels are shown: $b_i = \pm 0.2$, representing higher-accuracy AI, and $b_i = \pm 0.4$, representing lower-accuracy AI. Collective accuracy (A,C,E,G) and median reliance on AI (B,D,F,H) are averaged over generations 190,000–200,000. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 100^2)$ for (A–D), $c_A \sim \mathcal{N}(0, 5^2)$ for (E–H), and $\beta_A \sim U(0, 1)$. AI parameter: $\tau = 0.3$.

incentives or penalties can help the emergence of CI, we modify the two incentive structures by adding a payoff term proportional to individual reliance on AI:

$$\text{Feedback/Niche-expert} + \lambda\beta_A \quad (1)$$

where λ represent the strength of intervention ($\lambda > 0$) or penalty ($\lambda < 0$). [SWP: maybe we need to call incentive pi earlier on...] [SWP: also good to define beta again here as a reminder.]

Directly incentivizing or penalizing AI use can further reduce the collective accuracy (Fig. 3C, E, G), except when Omniscient AI is used under the feedback structure; in the latter case, the population retains collective intelligence regardless of the intervention strength (Fig. 3A). The reduction in collective accuracy arises because incentivizing or penalizing AI usage causes individuals to imitate others based on their AI usage rather than their predictive accuracy. Thus, stronger interventions

188 can push the group to always ($\beta = 1$) or never ($\beta = 0$) rely on AI (Fig. 3B,D, F, H)
 189 but these scenarios do not necessarily lead to better outcomes (Fig. 3D, F, H). When
 190 AI bias is relatively large, a weak penalty can lead to higher collective accuracy, but
 191 this represents a narrow parameter regime (Figure 3G).

192 Overall, these results indicate that simply incentivizing or penalizing the AI, in-
 193 dependent of individuals’ performance, use may exacerbate rather than resolve the
 194 risk of AI on CI. Thus, we also considered an incentive structure where interven-
 195 tions for AI usage affects the payoffs multiplicatively, thus giving higher payoffs to
 196 individuals who rely more (or less) AI and makes better prediction. [SWP: Supp.]

197 **Balanced incentives mitigate AI-induced failures of CI**

198 Neither the original incentive structures nor simple interventions targeting AI use are
 199 sufficient to promote CI. What incentive structure, then, would work in the presence
 200 of AI? Here, we propose a “balanced” incentive structure, modeled as the average of
 201 the feedback and niche expert incentives.

202 The core idea behind the balanced structure is to maintain interest diversity and
 203 prevent collective bias. On one hand, the feedback incentive structure is prone to
 204 the loss of interest diversity but can prevent collective bias by incentivizing players
 205 whose predictions lie opposite to the collective prediction. On the other hand, the
 206 niche-expert structure can maintain interest diversity because it directly incentivizes
 207 minority interests. However the niche-expert structure cannot prevent collective bias
 208 because, within the same factor of interest, individuals are rewarded only according
 209 to their individual predictive performance, which in turn homogenize the predictions.

210 Therefore, when both incentive structures are combined, the population achieves
 211 CI by maintaining both interest diversity and limited collective bias (Fig. 4, S3).
 212 Compare to the feedback and niche-expert incentive structures, individuals are both
 213 evenly distributed through factors (Fig. 4C–E), and their beliefs lie closer to true
 214 values (Fig. 4F–H). Moreover, the balanced incentive structure allows for higher lev-
 215 els of counterfactual collective accuracy compared to pure feedback and niche-expert
 216 scenarios. These patterns are driven by the emergent property of the balanced struc-
 217 ture, in which group increases its reliance on AI as the AI becomes more accurate,
 218 demonstrating appropriately calibrated AI use.

219 [SWP: *may need some robustness check rather than a case study...*]

220 **Discussion**

221 The ongoing race towards the development of “better” AIs continues to pose ques-
 222 tions about the societal benefits and risks of AI uses. Here, we used mathematical
 223 models of evolutionary dynamics to investigate how AI uses and incentive structures
 224 jointly affect the emergence of CI, which is defined as the population’s ability to make
 225 accurate predictions. We find that an “omniscient” AI model, which incorporates in-
 226 formation about all relevant causal factors, can improve collective accuracy, whereas

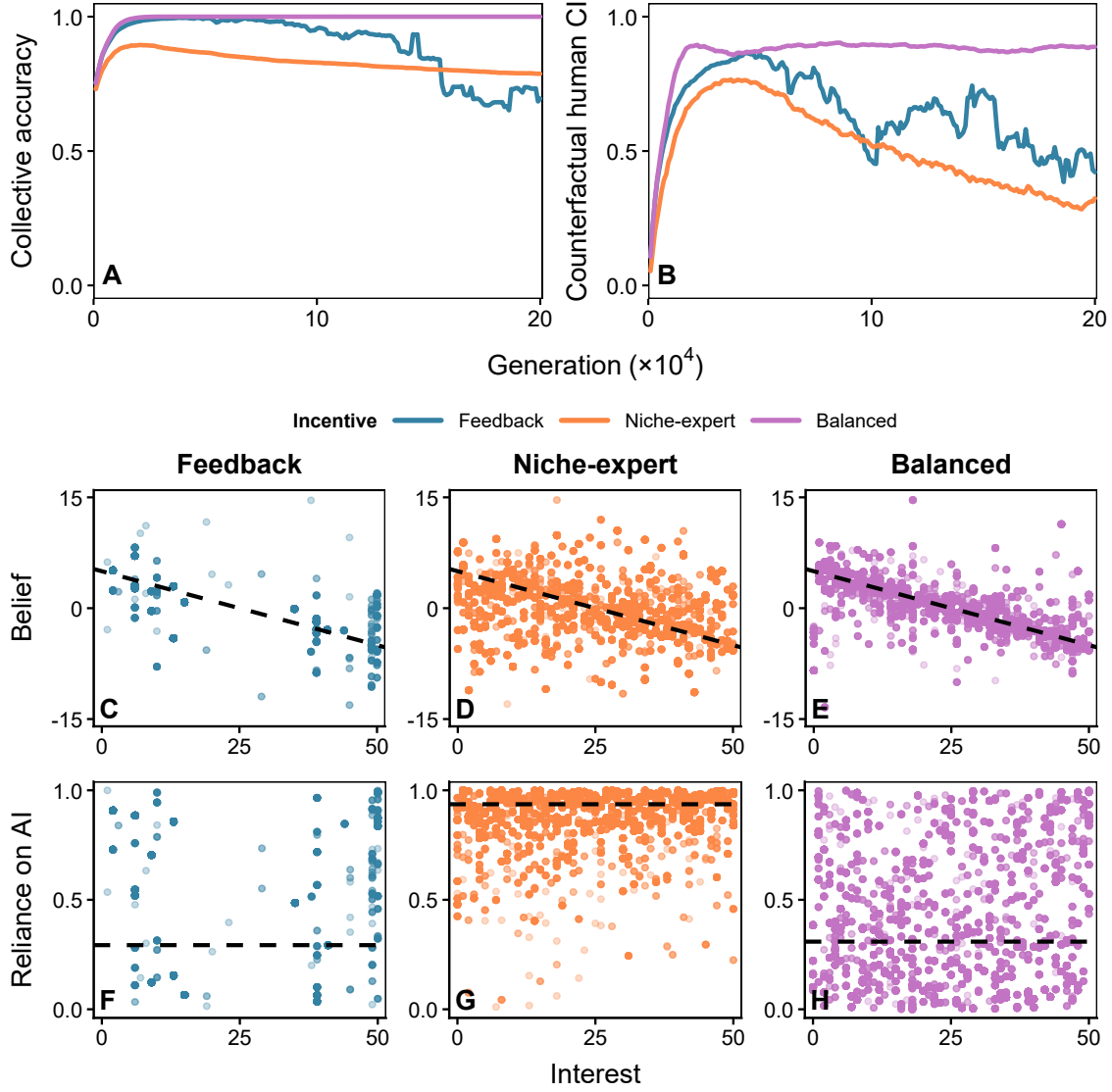


Figure 4: **Balanced structure can promote collective intelligence** Trajectories of collective accuracy (A) and counterfactual human CI (B) under three incentive structures (feedback, niche-expert, and balanced) with Chatbot AI. *[SWP: A and B should have the same y axes scaling.]* Distributions of individual interest and belief (C-E) and of interest and reliance on AI (F-H) after 200,000 generations. Each point represents an individual. Dashed lines in panels C-E represent the true coefficients (α_i) of the prediction target. Dashed horizontal lines in panels F-H indicate the median reliance on AI. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i = 5 - 0.2i$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 5^2)$, and $\beta_A \sim U(0, 1)$. AI parameters: $b_i = 0.4$ and $\tau = 0.3$.

227 a “chatbot” AI model, whose outcome depends on the user’s interest, can impair it.
228 These contrasting outcomes arise because AI assistance affects not only individual
229 predictions, but also the population’s interest diversity, beliefs, and reliance on AI.
230 We also show that directly incentivizing or penalizing AI uses, independent of in-
231 dividual’s performances, can further limit the collective accuracy of the population.
232 Instead, a balanced incentive structure can maintain CI by promoting diversity and
233 limiting bias.

234 Our analysis reveals that AI can prevent the emergence of CI through two dif-
235 ferent mechanisms. First, AI use can cause individuals to concentrate on a subset of
236 the factors that determine the prediction target, thereby reducing diversity in their
237 interests. This result is broadly consistent with recent empirical studies showing that
238 AI assistance can improve individual performance but limit the diversity of outputs,
239 in domains ranging from scientific research [31] to creative writing [17]. Second, over-
240 reliance on AI can cause the population to inherit inherent biases in AI’s predictions,
241 causing the prediction accuracy to decrease eventually. This mechanism can prevent
242 the emergence of CI even when the interest diversity is maintained because biases in
243 AI predictions can be amplified when individual predictions are aggregated. Thus,
244 preserving the diversity of ideas will not be sufficient to promote CI. Instead, AI
245 uses need to be calibrated to prevent systematic AI biases from becoming collective
246 biases.

247 Our analyses of counterfactual collective accuracy also raises an important con-
248 cern for future AI uses: while AI uses may improve the apparent performance of
249 individuals and, in some cases, the population as a whole, it can simultaneously re-
250 duce the population’s ability to make accurate predictions without AI. These results
251 echo findings from a recent field experiment, which demonstrated that students, who
252 seemingly performed better in solving mathematics problems under AI assistance,
253 performed worse on their own than those who did not have access [32]. These findings
254 highlight the need to evaluate AI tools not only by the performance gains they pro-
255 vide during use, but also by whether individuals and populations retain the ability to
256 perform without them. Making this distinction clear will be particularly important
257 for education, where AI may help students complete a task without helping them
258 learn how to do it on their own.

259 As generative AIs and large language models are already generating substantial
260 benefits, many organizations, both industrial and academic, are increasingly seeking
261 to improve their collective performance by regulating AI uses. Such regulations may
262 take direct forms, including AI-use policies or token-based leaderboards, or may op-
263 erate indirectly through other incentives, such as productivity rewards or the social
264 stigma associated with AI use. However, our model suggests that directly incentiviz-
265 ing or penalizing AI use is unlikely to induce appropriately calibrated reliance on
266 AI or to promote CI. Incentivizing AI use could lead to overreliance, causing groups
267 to propagate AI biases, whereas penalizing AI use could lead to underreliance, pre-
268 venting groups from using AI even when it performs well in the relevant domain.
269 Our result instead shows that appropriate AI use and accurate CI can be promoted

by focusing on the traditional properties of groups that have been emphasized for decades: maintaining diversity and preventing collective bias, rather than on AI use itself.

[SWP: need to be edited:] There are several limitations to our analyses. First, the model considers only social learning among humans and treats the influence of AI as unidirectional. In reality, humans can learn from AI, while AI systems can also be further trained on human-generated data. Such bidirectional adaptation may further reduce group performance by increasing the similarity between human and AI predictions and thereby weakening their complementarity. Empirical studies of human–AI collaboration likewise suggest that stronger correlations between human and AI predictions can diminish the benefits of collaboration. Second, we model AI errors using factor-specific additive biases and random error. Previous studies of AI bias and error suggest that additive biases may arise from skewed training data, or systematic calibration. However, AI errors may also exhibit more complex structures, including biased estimates of factor effects and correlations across prediction errors. Further empirical research is therefore needed to characterize the structure of AI bias and error, together with modeling approaches that can incorporate these more complex patterns. Finally, to retain the tractability of established CI models, we assume that the prediction target is a linear combination of causal factors, which does not hold for many real-world tasks. In particular, many machine learning methods used in AI systems are capable of capturing nonlinear relationships. Future research should therefore investigate how nonlinear relationships between causal factors and prediction targets affect the interaction between AI assistance and the emergence of CI.

Methods

Prediction target

Following previous models of CI, we model the prediction target as a linear combination of M independent causal factors:

$$Y = \alpha_0 + \alpha_1 X_1 + \cdots + \alpha_M X_M, \quad (2)$$

where the fixed coefficient α_i determines the magnitude and direction of the effect of factor i , and α_0 is a global intercept. Each causal factor X_i is modeled as an independent Gaussian random variable with mean zero and standard deviation σ_i .

Players

We consider a well-mixed population $\mathcal{N}$ consisting of N players engaged in a novel prediction task. In the absence of AI assistance, the strategy of player A is characterized by two variables: interest e_A and belief c_A . Player A observes the causal

305 factor X_{e_A} and predicts the target by multiplying the observed factor value by their
 306 belief:

$$Y_A = c_A X_{e_A}, \quad (3)$$

307 where e_A can take an integer value from 0 to M . When $e_A = 0$, the player
 308 does not observe any causal factor and instead makes a constant prediction based
 309 solely on their belief. We refer to this as a stubborn strategy because the player's
 310 prediction does not respond to the realized values of the causal factors. For notational
 311 convenience, we assume that $X_0 = 1$.

312 When the group is assisted by AI, each player's strategy additionally includes
 313 reliance on AI, denoted by β_A . We assume that each player consults the AI inde-
 314 pendently and revises their prediction by taking a weighted average of their own
 315 prediction and the AI-generated prediction:

$$Y_A^{\text{re}} = \beta_A Y_A^{\text{AI}} + (1 - \beta_A) Y_A, \quad (4)$$

316 where β_A ranges from 0 to 1. A value of $\beta_A = 1$ indicates complete reliance on
 317 AI, whereas $\beta_A = 0$ indicates that the player does not use the AI prediction.

318 AI models

319 We assume that the AI model has been trained on similar tasks and is therefore
 320 capable of making predictions based on causal factors. However, because the model
 321 is imperfect, its outputs are subject to systematic additive bias and stochastic error.
 322 The additive bias associated with factor i is denoted by b_i , and the stochastic error
 323 is denoted by ϵ . The AI is neither trained nor updated during the simulation, and its
 324 biases therefore remain fixed throughout the simulation. Although the bias is indexed
 325 by factor, all b_i , including b_0 , are assigned the same value in the main analyses for
 326 simplicity. The model formulation nevertheless allows biases to differ among factors.

327 We assume that the stochastic error follows a Gaussian distribution with mean
 328 zero and standard deviation τ . Each time a player consults the AI, an indepen-
 329 dent error is generated. Errors are therefore independent across players and across
 330 repeated consultations by the same player. Consequently, stochastic AI errors can
 331 be attenuated when independently generated predictions are aggregated across play-
 332 ers. We therefore held τ constant and varied the magnitude of the additive bias to
 333 manipulate the level of AI accuracy across analyses.

334 We consider two extreme types of AI that differ in the amount of information
 335 available for prediction. The first is termed the Omniscient AI, which can observe
 336 all M causal factors and incorporate them into its prediction regardless of the user's
 337 interest. The prediction generated by the Omniscient AI for player A is expressed
 338 as follows:

$$Y_A^{\text{AI}} = \alpha_0 + b_0 + \alpha_1 X_1 + b_1 + \cdots + \alpha_M X_M + b_M + \epsilon. \quad (5)$$

339 The second is termed the Chatbot AI, which can use only the single causal factor
 340 X_{e_A} associated with the player’s interest. The prediction generated by the Chatbot
 341 AI for player A is expressed as follows:

$$Y_A^{\text{AI}} = \alpha_{e_A} X_{e_A} + b_{e_A} + \epsilon. \quad (6)$$

342 Aggregation rules

343 The revised individual predictions are aggregated to form a collective prediction. We
 344 use two aggregation rules proposed in previous research, depending on the type of
 345 AI used by the group.

346 When the group is assisted by the Omniscient AI, the AI component of each
 347 revised prediction contains information about all M factors, even though the indi-
 348 vidual’s own prediction is based on only one factor. Each revised prediction can
 349 therefore be treated as an estimate of the entire target. Accordingly, we calculate
 350 the collective prediction by taking the simple average of the revised individual pre-
 351 dictions:

$$\hat{Y}^{\text{re}} = \frac{Y_1^{\text{re}} + Y_2^{\text{re}} + \dots + Y_n^{\text{re}}}{N}. \quad (7)$$

352 When the group is assisted by the Chatbot AI, both the player’s initial prediction
 353 and the AI’s prediction are based on the same causal factor. Each revised predic-
 354 tion therefore contains information about only one factor and can be treated as an
 355 estimate of that factor’s contribution to the target. We consequently use clustering
 356 aggregation.

357 Clustering aggregation is performed by grouping individuals according to their
 358 interests and calculating the mean prediction within each group. The collective
 359 prediction is then obtained by summing the mean predictions of all nonempty groups:

$$\hat{Y}^{\text{re}} = \sum_{i:n_i>0} \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} Y_A^{\text{re}}, \quad (8)$$

360 where $\mathcal{A}_i$ is the set of players interested in factor i , and $n_i = |\mathcal{A}_i|$ is the number
 361 of players in that set.

362 When the group is not assisted by AI, collective predictions are computed from the
 363 unrevised individual predictions throughout the simulation. Both simple averaging
 364 and clustering can be used in the absence of AI. For fair comparison, we use the
 365 same aggregation rule as in the corresponding AI-assisted condition.

366 Incentive structures

367 Players receive payoffs determined by an incentive structure based on their strategies.
 368 We consider five incentive structures: feedback structure, niche-expert structure,
 369 direct intervention on AI use of these two structures, and a balanced structure.

370 The feedback structure does not directly reward individuals according to their
 371 own prediction accuracy. Instead, it rewards predictions that are aligned with the
 372 residual error of the collective prediction and therefore have the potential to move
 373 the collective prediction toward the true value. Given player A 's revised prediction
 374 Y_A^{re} and the collective prediction $\hat{Y}^{\text{re}}$, the payoff is expressed as

$$\pi_A = Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}). \quad (9)$$

375 The niche-expert structure assigns higher payoffs to individuals who make accu-
 376 rate predictions while having an underrepresented interest. Specifically, the squared
 377 prediction error of player A is weighted by $\rho_{e_A} = n_{e_A}/n$, the proportion of players
 378 interested in the same factor:

$$\pi_A = -\rho_{e_A}(Y_A^{\text{re}} - Y)^2. \quad (10)$$

379 Previous research has shown that these two incentive structures can produce
 380 nearly perfect CI through social learning, except when the niche-expert incentive is
 381 combined with simple averaging. They are therefore used as baseline structures for
 382 examining how AI assistance affects the emergence of CI.

383 To examine the effects of directly encouraging or discouraging AI use, we mod-
 384 ify the feedback and niche-expert structures by adding a term proportional to the
 385 player's reliance on AI:

$$\pi_A = Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) + \lambda\beta_A. \quad (11)$$

$$\pi_A = -\rho_{e_A}(Y_A^{\text{re}} - Y)^2 + \lambda\beta_A. \quad (12)$$

386 Here, λ is a weighting coefficient representing the strength of the direct incentive
 387 or penalty applied to AI use. A positive value of λ rewards AI reliance, whereas a
 388 negative value penalizes it. When $\lambda = 0$, no direct incentive or penalty is applied,
 389 and the payoff functions reduce to the original feedback and niche-expert structures.

390 The balanced structure is defined as the arithmetic mean of the feedback and
 391 niche-expert payoff components:

$$\pi_A = \frac{Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) - \rho_{e_A}(Y_A^{\text{re}} - Y)^2}{2}. \quad (13)$$

392 Simulations

393 All simulations were conducted using an agent-based model with 50 causal factors
 394 ($M = 50$) and 10,000 players ($n = 10,000$). The parameters defining the prediction
 395 target were sampled at the beginning of the simulations and subsequently held fixed.

396 The coefficients α_i were sampled from a uniform distribution on $[-5, 5]$, except in
 397 the simulation presented in Figure 4, in which α_i was set to $5 - 0.2i$ for visualization.
 398 The standard deviations σ_i were sampled from a uniform distribution on $[0, 3]$.

399 Players’ initial strategies were also sampled from predefined probability distri-
400 butions. The distribution of individual belief c_A depended on the aggregation rule.
401 Under simple averaging, each factor’s contribution is scaled by the proportion of
402 players interested in that factor. We therefore sampled beliefs from a Gaussian dis-
403 tribution with mean zero and standard deviation 100, following the scaling used
404 in previous work [13]. Under clustering aggregation, beliefs were sampled from a
405 Gaussian distribution with mean zero and standard deviation 5. Because predictions
406 are averaged separately within each interest group and then summed, no additional
407 scaling of c_A is required. The initial interest of each player was sampled uniformly
408 from the integers between 0 and 50, and reliance on AI was sampled from a uniform
409 distribution on $[0, 1]$.

410 We assumed that prediction events occurred on a substantially faster timescale
411 than social learning. At each generation, we therefore calculated the expected payoff
412 $\mathbb{E}[\pi_A]$ of each player analytically rather than explicitly sampling prediction events.
413 The expectation was taken over the distributions of the causal factors $X_1, \dots, X_M$
414 and the stochastic AI error ϵ , conditional on the current population strategies and
415 the fixed target parameters. Closed form expression of expected payoffs were derived
416 for all incentive structures and are presented in the Supplementary Methods.

417 Each generation consisted of an expected payoff calculation followed by one op-
418 portunity for pairwise social learning. Two distinct players, A and B , were randomly
419 sampled, and player A imitated all three strategies of player B with probability

$$\frac{1}{1 + \exp(s(\mathbb{E}[\pi_A] - \mathbb{E}[\pi_B]))}, \quad (14)$$

420 where s represents the strength of social learning and was set to 50 in all sim-
421 ulations. We introduced a mutation rate μ , such that with probability μ , player A
422 did not imitate player B but instead redrew all strategy components from the same
423 distributions used to initialize the population. The mutation rate was set to zero in
424 all simulations except those presented in Supplementary Figure 4.

425 Unless otherwise specified, each simulation was run for 200,000 generations. When
426 a result was represented by a single summary value rather than as a temporal tra-
427 jectory, the metric was averaged over the final 10,000 generations.

428 To evaluate the dynamics of collective prediction, we defined six summary met-
429 rics: collective accuracy, collective bias, collective variance, counterfactual collective
430 accuracy, interest diversity, and median reliance on AI. At each population state,
431 expectations and variances were evaluated analytically over the distributions of the
432 causal factors and AI errors rather than estimated from sampled prediction events.
433 Collective bias and collective variance constitute the bias–variance decomposition of
434 the group level collective prediction error.

435 Collective accuracy was normalized relative to the baseline prediction α_0 . A neg-
436 ative value indicates that the collective prediction performs worse than the baseline.
437 Counterfactual collective accuracy was evaluated by setting $\beta_A = 0$ while retaining
438 the evolved interests and beliefs of the AI-assisted population. In calculating interest

439 diversity, we used the convention that $\rho_i \ln \rho_i = 0$ when $\rho_i = 0$ (Table 1).

Table 1: Evaluation metrics used to analyze collective prediction dynamics.

Metric	Definition	Interpretation
Collective accuracy	$1 - \frac{\mathbb{E}[(Y - \hat{Y}^{\text{re}})^2]}{\mathbb{E}[(Y - \alpha_0)^2]}$	Accuracy of the revised collective prediction relative to the baseline prediction α_0 .
Collective bias	$\left(\mathbb{E}[Y - \hat{Y}^{\text{re}}]\right)^2$	Squared expected deviation of the revised collective prediction from the true target.
Collective variance	$\text{Var}(Y - \hat{Y}^{\text{re}})$	Variability of the revised collective prediction error across prediction events.
Counterfactual collective accuracy	$1 - \frac{\mathbb{E}[(Y - \hat{Y})^2]}{\mathbb{E}[(Y - \alpha_0)^2]}$	Accuracy obtained after removing AI use from the current population while retaining its evolved interests and beliefs.
Interest diversity	$\exp\left(-\sum_{i=0}^M \rho_i \ln \rho_i\right)$	Effective number of interests represented in the population.
Median reliance on AI	$\text{median}_{A \in N}(\beta_A)$	Median reliance on AI in the population.

440 Data and code availability

441 All data are stored in a publicly available GitHub repository ([https://github.com/](https://github.com/parklab-snu/collective-intelligence-with-AI)
442 [parklab-snu/collective-intelligence-with-AI](https://github.com/parklab-snu/collective-intelligence-with-AI)).

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447 **Contributions**

448 Jaeseung Kim: Conceptualization, Formal Analysis, Investigation, Methodology,
449 Software, Visualization, Writing - Original Draft Preparation, Writing - Review &
450 Editing. Sang Woo Park: Conceptualization, Investigation, Methodology, Supervi-
451 sion, Writing - Original Draft Preparation, Writing - Review & Editing.

452 **Competing Interests**

453 The authors declare no competing interests.

454 Supplementary Text

455 1 Omniscient AI

456 Omniscient AI assumes that AI can incorporate all causal factors regardless of the
457 player's interest.

458 1.1 Model

$$Y_A^{AI} = \alpha_0 + b_0 + \alpha_1 X_1 + b_1 + \cdots + \alpha_M X_M + b_M + \epsilon. \quad (S1)$$

459 For simplicity, prediction of Omniscient AI can be written as:

$$Y^{AI} = Y + B + \epsilon \quad (S2)$$

460 where Y stands for true prediction target ($Y = \alpha_0 + \alpha_1 X_1 + \cdots + \alpha_M X_M$) and
461 B is sum of all fixed biases ($B = b_0 + b_1 + \cdots + b_M$), ϵ follows normal distribution
462 with zero mean, τ standard deviation with no correlation within or between players.
463 Player A's prediction and revised prediction can be written as follows:

$$Y_A = c_A X_{e_A} \quad (S3)$$

464

$$Y_A^{re} = \beta_A Y^{AI} + (1 - \beta_A) Y_A \quad (S4)$$

465 where c_A stands for player A's belief on the factor, e_A stands for the interest, and
466 β_A stands for reliance on AI.

467 1.2 Aggregation

468 Because Y_A^{AI} is subject to all M factors, we use simple averaging aggregation for
469 collective prediction.

$$\begin{aligned} \hat{Y}^{re} &= \frac{1}{N} \sum_{j=1}^N Y_j^{re} \\ &= \frac{1}{N} \sum_{j=1}^N \{ \beta_j Y^{AI} + (1 - \beta_j) Y_j \} \\ &= \frac{1}{N} \sum_{j=1}^N \{ \beta_j (Y + B + \epsilon) + (1 - \beta_j) c_j X_{e_j} \} \end{aligned}$$

470 Reparameterization for simple derivation and interpretability.

$$\bar{\beta} := \frac{1}{N} \sum_{j=1}^N \beta_j \quad (S5)$$

471

$$\bar{\eta} := \frac{1}{N} \sum_{j=1}^N \beta_j \epsilon \quad (\text{S6})$$

472

$$\gamma_i := \frac{1}{N} \sum_{A \in \mathcal{A}_i} (1 - \beta_A) c_A \quad (\text{S7})$$

473 $\bar{\beta}$ can be interpreted as mean reliance on AI, γ stands for the coefficient for the factor
 474 i which is solely based on players' belief.

475 By this reparameterization, we can write $\hat{Y}^{re}$ as follows.

$$\hat{Y}^{re} = \bar{\beta}(Y + B) + \sum_{i=0}^M \gamma_i X_i + \bar{\eta} \quad (\text{S8})$$

476 1.3 Expected payoffs

477 Feedback

$$\delta_0 := \alpha_0 - (\bar{\beta}(\alpha_0 + B) + \gamma_0) \quad (\text{S9})$$

478

$$\delta_i := \alpha_i - (\bar{\beta}\alpha_i + \gamma_i), \quad (i \neq 0) \quad (\text{S10})$$

479

$$Y - \hat{Y}^{re} = \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta} \quad (\text{S11})$$

480 See equation Eq. (S6) for the $\bar{\eta}$.

481 δ_0 is constant term and δ_i is independent of α_k ($k \neq i$), ϵ , and $\bar{\eta}$.

$$\begin{aligned} \mathbb{E}(\pi_A) &= \mathbb{E}[Y_A^{re}(Y - \hat{Y}^{re})] \\ &= \mathbb{E} \left[\left\{ \beta_A \alpha_0 + \beta_A \sum_{i=1}^M \alpha_i X_i + \beta_A B + \beta_A \epsilon + (1 - \beta_A) c_A X_{e_A} \right\} \left\{ \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta} \right\} \right] \\ &= \beta_A \alpha_0 \delta_0 + \beta_A \sum_{i=1}^M \delta_i \alpha_i \sigma_i^2 + \beta_A B \delta_0 - \frac{1}{N} \beta_A^2 \tau^2 + (1 - \beta_A) c_A \delta_{e_A} \sigma_{e_A}^2 \\ &= \beta_A \left\{ (\alpha_0 + B) \delta_0 + \sum_{i=1}^M \alpha_i \delta_i \sigma_i^2 \right\} + (1 - \beta_A) c_A \delta_{e_A} \sigma_{e_A}^2 - \frac{\beta_A^2 \tau^2}{N} \end{aligned}$$

482 **Niche-expert**

483 If $e_A = 0$, player does not observe causal factor and their unrevised prediction is same
 484 with their belief c_A . Such that, the expected payoff is

$$\begin{aligned}
 \mathbb{E}(\pi_A) &= -\mathbb{E}[\rho_0(Y_A^{re} - Y)^2] \\
 &= -\mathbb{E}[\rho_0\{(\beta_A - 1)Y + \beta_A B + \beta_A \epsilon + (1 - \beta_A)c_A\}^2] \\
 &= -\mathbb{E}\left[\rho_0\left\{(\beta_A - 1)(\alpha_0 - c_A) + \beta_A B + (\beta_A - 1)\sum_{i=1}^M \alpha_i X_i + \beta_A \epsilon\right\}^2\right] \\
 &= -\rho_0\left\{(1 - \beta_A)^2(\alpha_0 - c_A)^2 - 2\beta_A(1 - \beta_A)(\alpha_0 - c_A)B + \beta_A^2 B^2\right. \\
 &\quad \left.+ (1 - \beta_A)^2 \sum_{i \neq 0} \alpha_i^2 \sigma_i^2 + \beta_A^2 \tau^2\right\}
 \end{aligned}$$

485 Else,

$$\begin{aligned}
 \mathbb{E}(\pi_A) &= -\mathbb{E}[\rho_{e_A}(Y_A^{re} - Y)^2] \\
 &= -\mathbb{E}[\rho_{e_A}\{(\beta_A - 1)Y + \beta_A B + \beta_A \epsilon + (1 - \beta_A)c_A X_{e_A}\}^2] \\
 &= -\mathbb{E}\left[\rho_{e_A}\left\{(\beta_A - 1)\alpha_0 + \beta_A B + (\beta_A - 1)(\alpha_{e_A} - c_A)X_{e_A} + (\beta_A - 1)\sum_{i \neq e_A, 0} \alpha_i X_i + \beta_A \epsilon\right\}^2\right] \\
 &= -\rho_{e_A}\left\{(1 - \beta_A)^2 \alpha_0^2 - 2\beta_A(1 - \beta_A)\alpha_0 B + \beta_A^2 B^2 + (1 - \beta_A)^2(\alpha_{e_A} - c_A)^2 \sigma_{e_A}^2\right. \\
 &\quad \left.+ (1 - \beta_A)^2 \sum_{i \neq e_A, 0} \alpha_i^2 \sigma_i^2 + \beta_A^2 \tau^2\right\}
 \end{aligned}$$

486 **Incentivizing or penalizing AI use**

487 Because additive term $\lambda\beta$ is not a random variable, we can simply derive the expected
 488 payoff as

$$\mathbb{E}(\pi_{A, Feedback}) = \mathbb{E}[Y_A^{re}(Y - \hat{Y}^{re}) + \lambda\beta_A] = \mathbb{E}[Y_A^{re}(Y - \hat{Y}^{re})] + \lambda\beta_A. \quad (\text{S12})$$

$$\mathbb{E}(\pi_{A, Niche-expert}) = -\mathbb{E}[\rho_{e_A}(Y_A^{re} - Y)^2 + \lambda\beta_A] = -\mathbb{E}[\rho_{e_A}(Y_A^{re} - Y)^2] + \lambda\beta_A. \quad (\text{S13})$$

489 **1.4 Expected accuracy**

$$\mathbb{E}[(Y - \alpha_0)^2] = \mathbb{E}\left[\left(\alpha_0 + \sum_{i=1}^M \alpha_i X_i - \alpha_0\right)^2\right] = \sum_{i=1}^M \alpha_i^2 \sigma_i^2 \quad (\text{S14})$$

490

$$\mathbb{E}[(Y - \hat{Y}^{re})^2] = \mathbb{E}\left[\left(\delta_0 + \sum_{i=1}^M \delta_i X_i - \hat{\eta}\right)^2\right] = \delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \frac{\tau^2}{N^2} \sum_{j=1}^N \beta_j^2 \quad (\text{S15})$$

491

$$\begin{aligned} R &= 1 - \frac{\mathbb{E}[(Y - \hat{Y})^2]}{\mathbb{E}[(Y - \alpha_0)^2]} \\ &= 1 - \frac{\delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \frac{\tau^2}{N^2} \sum_{j=1}^N \beta_j^2}{\sum_{i=1}^M \alpha_i^2 \sigma_i^2} \end{aligned}$$

492 2 Chatbot AI

493 2.1 Model

$$Y_A^{AI} = \alpha_{e_A} X_{e_A} + b_{e_A} + \epsilon \quad (\text{S16})$$

494 b_{e_A} stands for factor specific additive bias, ϵ follows normal distribution with zero
 495 mean, τ standard deviation with no correlation within or between players. Player
 496 A's prediction and revised prediction can be written as follows:

$$Y_A = c_A X_{e_A} \quad (\text{S17})$$

497

$$Y_A^{re} = \beta_A Y_A^{AI} + (1 - \beta_A) Y_A \quad (\text{S18})$$

498 where c_A stands for player A's belief on the factor, e_A stands for the interest, and
 499 β_A stands for reliance on AI.

500 2.2 Aggregation

501 For Chatbot AI, clustering aggregation is used for collective prediction.

$$\begin{aligned} \hat{Y}^{re} &= \sum_{i=0}^M \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} Y_A^{re} \\ &= \sum_{i=0}^M \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \{\beta_A Y_A^{AI} + (1 - \beta_A) Y_A\} \\ &= \sum_{i=0}^M \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} [\{\beta_A \alpha_{e_A} + (1 - \beta_A) c_A\} X_{e_A} + \beta_A b_{e_A} + \beta_A \epsilon] \end{aligned}$$

502 Reparameterize for the simplicity.

$$\tilde{c}_i := \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \{\beta_A \alpha_{e_A} + (1 - \beta_A) c_A\} \quad (\text{S19})$$

503

$$\tilde{B}_i := \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \beta_A b_{e_A} \quad (\text{S20})$$

504

$$\bar{B} := \sum_{i=0}^M \tilde{B}_i \quad (\text{S21})$$

505

$$\tilde{\eta}_i := \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \beta_A \epsilon \quad (\text{S22})$$

506

$$\bar{\eta} := \sum_{i=0}^M \tilde{\eta}_i \quad (\text{S23})$$

507 2.3 Expected payoffs

508 Feedback

509

$$\delta_0 := \alpha_0 - \tilde{c}_0 - \bar{B} \quad (\text{S24})$$

510

$$\delta_i := \alpha_i - \tilde{c}_i, \quad (i \neq 0) \quad (\text{S25})$$

$$Y - \hat{Y}^{re} = \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta} \quad (\text{S26})$$

511 δ_0 is constant term and δ_i is independent of α_k ($k \neq i$), ϵ , and $\bar{\eta}$.

$$\begin{aligned} \mathbb{E}(\pi_A) &= \mathbb{E}[Y_A^{re}(Y - \hat{Y}^{re})] \\ &= \mathbb{E} \left[\{(\beta_A \alpha_{e_A} + (1 - \beta_A) c_A) X_{e_A} + \beta_A b_{e_A} + \beta_A \epsilon\} \left\{ \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta} \right\} \right] \\ &= \{\beta_A \alpha_{e_A} + (1 - \beta_A) c_A\} \delta_{e_A} \sigma_{e_A}^2 + \beta_A b_{e_A} \delta_0 - \frac{1}{n_{e_A}} \beta_A^2 \tau^2 \end{aligned}$$

512 Niche-expert

513 If $e_A = 0$,

$$\begin{aligned} \mathbb{E}(\pi_A) &= -\mathbb{E}[\rho_0(Y_A^{re} - Y)^2] \\ &= -\mathbb{E} \left[\rho_0 \left[\{\beta_A \alpha_0 + (1 - \beta_A) c_A\} + \beta_A b_0 + \beta_A \epsilon - \alpha_0 - \sum_{i=1}^M \alpha_i X_i \right]^2 \right] \\ &= -\rho_0 \left\{ (1 - \beta_A)^2 (c_A - \alpha_0)^2 + 2(1 - \beta_A)(c_A - \alpha_0) \beta_A b_0 + \beta_A^2 b_0^2 \right. \\ &\quad \left. + \sum_{i \neq 0} \alpha_i^2 \sigma_i^2 + \beta_A^2 \tau^2 \right\} \end{aligned}$$

514 Else,

$$\begin{aligned}
\mathbb{E}(\pi_A) &= -\mathbb{E}[\rho_{e_A}(Y_A^{\text{re}} - Y)^2] \\
&= -\mathbb{E}\left[\rho_{e_A}\left[\{\beta_A\alpha_{e_A} + (1 - \beta_A)c_A\}X_{e_A} + \beta_A b_{e_A} + \beta_A\epsilon - \alpha_0 - \sum_{i=1}^M \alpha_i X_i\right]^2\right] \\
&= -\rho_{e_A}\left\{(1 - \beta_A)^2(c_A - \alpha_{e_A})^2\sigma_{e_A}^2 + \sum_{i \neq e_A, 0} \alpha_i^2 \sigma_i^2\right. \\
&\quad \left.+ \alpha_0^2 + \beta_A^2 b_{e_A}^2 - 2\alpha_0\beta_A b_{e_A} + \beta_A^2 \tau^2\right\}
\end{aligned}$$

515 Incentivizing or penalizing AI use

516 Because additive term $\lambda\beta$ is not a random variable, we can simply derive the expected
517 payoff as

$$\mathbb{E}(\pi_{A, \text{Feedback}}) = \mathbb{E}\left[Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) + \lambda\beta_A\right] = \mathbb{E}\left[Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}})\right] + \lambda\beta_A. \quad (\text{S27})$$

$$\mathbb{E}(\pi_{A, \text{Niche-expert}}) = -\mathbb{E}\left[\rho_{e_A}(Y_A^{\text{re}} - Y)^2 + \lambda\beta_A\right] = -\mathbb{E}\left[\rho_{e_A}(Y_A^{\text{re}} - Y)^2\right] + \lambda\beta_A. \quad (\text{S28})$$

518 Balanced

519 Expected payoff of the balanced structure can also be driven by arithmetic mean of
520 expected payoff of feedback and niche-expert.

$$\mathbb{E}(\pi_A) = \mathbb{E}\left[\frac{Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) - \rho_{e_A}(Y_A^{\text{re}} - Y)^2}{2}\right] = \frac{\mathbb{E}\left[Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}})\right] - \mathbb{E}\left[\rho_{e_A}(Y_A^{\text{re}} - Y)^2\right]}{2} \quad (\text{S29})$$

521 2.4 Expected accuracy

$$\mathbb{E}[(Y - \alpha_0)^2] = \mathbb{E}\left[\left(\alpha_0 + \sum_{i=1}^M \alpha_i X_i - \alpha_0\right)^2\right] = \sum_{i=1}^M \alpha_i^2 \sigma_i^2 \quad (\text{S30})$$

522

$$\mathbb{E}[(Y - \hat{Y})^2] = \mathbb{E}\left[\left(\delta_0 + \sum_{i=1}^M \delta_i X_i - \hat{\eta}\right)^2\right] = \delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \sum_{i=0}^M \frac{\tau^2}{n_i^2} \sum_{A \in \mathcal{A}_i} \beta_A^2 \quad (\text{S31})$$

$$\begin{aligned}
R &= 1 - \frac{\mathbb{E}[(Y - \hat{Y})^2]}{\mathbb{E}[(Y - \alpha_0)^2]} \\
&= 1 - \frac{\delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \sum_{i=0}^M \frac{\tau^2}{n_i^2} \sum_{A \in \mathcal{A}_i} \beta_A^2}{\sum_{i=1}^M \alpha_i^2 \sigma_i^2}
\end{aligned}$$

524 3 Evolutionary dynamics

525 For sufficiently large population ($N \rightarrow \infty$), we can approximate the deterministic
526 evolutionary dynamics using the replicator equation. Let λ denote a strategy, p_λ its
527 frequency, and $u_\lambda = \mathbb{E}(\pi_\lambda)$ its expected payoff. The replicator equation is

$$\frac{dp_\lambda}{dt} = p_\lambda(u_\lambda - \bar{u}), \quad (\text{S32})$$

528 where $\bar{u} = \sum_\lambda p_\lambda u_\lambda$. Therefore, at an internal stationary state, all strategies
529 with positive frequency have the same expected payoff. A boundary state can also
530 be stationary because a strategy with exactly zero frequency cannot increase under
531 the replicator equation. Thus, stability of a boundary state needs to be considered
532 separately by examining invasion by a rare strategy.

533 3.1 Omniscient AI

534 3.1.1 Feedback

535 When the population size approaches infinity, the effect of stochastic error on the
536 collective prediction vanishes ($\bar{\eta} \rightarrow 0$) and the effect of stochastic error in the ex-
537 pected payoff, $-\beta^2 \tau^2 / N$, also vanishes. Following prior research [13], we define the
538 Lyapunov function as the expected error of collective prediction

$$V = \mathbb{E} \left[(Y - \hat{Y}^{re})^2 \right]. \quad (\text{S33})$$

539 The function V is non-negative and equals zero when the collective prediction is
540 identical to the target. To examine how V changes under the evolutionary dynamics,
541 we take its time derivative:

$$\begin{aligned}
\frac{dV}{dt} &= -2\mathbb{E} \left[(Y - \hat{Y}^{re}) \frac{d\hat{Y}^{re}}{dt} \right] \\
&= -2\mathbb{E} \left[\left(\delta_0 + \sum_{i=1}^M \delta_i X_i \right) \left\{ \frac{d\bar{\beta}}{dt} (Y + B) + \sum_{i=0}^M \frac{d\gamma_i}{dt} X_i \right\} \right] \\
&= -2 \left[\left\{ (\alpha_0 + B) \delta_0 + \sum_{i=1}^M \alpha_i \delta_i \sigma_i^2 \right\} \frac{d\bar{\beta}}{dt} + \delta_0 \frac{d\gamma_0}{dt} + \sum_{i=1}^M \delta_i \sigma_i^2 \frac{d\gamma_i}{dt} \right].
\end{aligned}$$

542 For notational convenience, we define

$$K = (\alpha_0 + B) \delta_0 + \sum_{i=1}^M \alpha_i \delta_i \sigma_i^2. \quad (\text{S34})$$

543 Although K changes with the population state, it is common to all strategies at a
544 given state and can therefore be treated as a constant with respect to the strategy
545 index λ . We also define

$$G_e = \begin{cases} \delta_0, & e = 0, \\ \delta_e \sigma_e^2, & e \neq 0. \end{cases} \quad (\text{S35})$$

546 By the definitions of $\bar{\beta}$ and γ_i and the replicator equation,

$$\frac{d\bar{\beta}}{dt} = \sum_{\lambda} p_{\lambda} (u_{\lambda} - \bar{u}) \beta_{\lambda} = \text{Cov}(\beta, u). \quad (\text{S36})$$

547 Similarly, the terms involving γ_i can be combined as

$$\delta_0 \frac{d\gamma_0}{dt} + \sum_{i=1}^M \delta_i \sigma_i^2 \frac{d\gamma_i}{dt} = \text{Cov}((1 - \beta) c G_e, u). \quad (\text{S37})$$

548 Therefore,

$$\begin{aligned}
\frac{dV}{dt} &= -2 \{ K \text{Cov}(\beta, u) + \text{Cov}((1 - \beta) c G_e, u) \} \\
&= -2 \text{Cov}(K\beta + (1 - \beta) c G_e, u).
\end{aligned}$$

549 For the feedback structure in the limit $N \rightarrow \infty$, the expected payoff of strategy
550 λ is

$$u_{\lambda} = K \beta_{\lambda} + (1 - \beta_{\lambda}) c_{\lambda} G_{e_{\lambda}}. \quad (\text{S38})$$

551 Which is same with Thus,

$$\frac{dV}{dt} = -2 \text{Cov}(u, u) = -2 \text{Var}(u) \leq 0. \quad (\text{S39})$$

Therefore, the expected error of collective prediction cannot increase during the evolutionary process and decreases whenever strategies differ in their expected payoffs. The derivative becomes zero when the variance of expected payoffs is zero, $\text{Var}(u) = 0$. This condition corresponds to equal expected payoffs among the strategies present in the population. A boundary state in which the entire population consists of a single strategy can also satisfy $\text{Var}(u) = 0$ even when $V > 0$. However, such a state is not evolutionarily stable under the feedback structure. If the collective prediction differs from the target Y , a nonzero residual remains. By the definition of the feedback payoff, a novel strategy whose prediction aligns with the direction of this residual obtains a higher expected payoff than the resident strategy and can therefore invade the population. Thus, an imperfect boundary equilibrium with $V > 0$ is unstable to the introduction of novel predictions.

Consequently, the evolutionary dynamics continue to reduce the collective prediction error until the population reaches the stable state with zero residual, $V = 0$. At this state, the collective prediction equals the target, $\hat{Y}^{re} = Y$, and the collective accuracy reaches $R = 1$.

3.1.2 Niche-expert

Unlike the stochastic error in the collective prediction, stochastic error in the niche-expert payoff is evaluated at the individual level and does not vanish as population size increases. Therefore, we retain $\tau > 0$ in the following derivation.

We first consider strategies having the same interest by optimizing the expected payoff with respect to c and β . For a nonzero interest e , the expected payoff is strictly concave with respect to c for $\beta < 1$ and $\sigma_e > 0$, and is maximized at

$$c_e^* = \alpha_e. \quad (\text{S40})$$

After substituting $c_e^* = \alpha_e$, the payoff is strictly concave with respect to β . Its unconstrained stationary point is

$$\tilde{\beta}_e = \frac{\alpha_0^2 + \alpha_0 B + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2}{(\alpha_0 + B)^2 + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S41})$$

Because β is constrained between 0 and 1, the optimal β_e^* is $\tilde{\beta}_e$ when $0 < \tilde{\beta}_e < 1$, and otherwise lies at the corresponding boundary, $\beta_e^* = 0$ or $\beta_e^* = 1$. Thus, the boundary cases do not require a separate evolutionary argument; they only change the optimal prediction and its minimum expected squared error. When $\beta_e^* = 1$, the revised prediction is independent of c_e , so different values of c_e are selectively neutral and can be treated as a single payoff-equivalent strategy class.

Let L_e denote the minimum expected individual error, $\mathbb{E}[(Y_A^{re} - Y)^2]$, within interest e . For an internal optimum,

$$L_e = \frac{B^2 \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 + \tau^2 \left(\alpha_0^2 + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 \right)}{(\alpha_0 + B)^2 + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S42})$$

585 If the optimum is located at $\beta_e^* = 0$ or $\beta_e^* = 1$, L_e is instead obtained by evaluating
 586 the expected squared prediction error at the corresponding boundary.

587 For interest 0, optimization with respect to c gives

$$c_0^*(\beta) = \alpha_0 - \frac{\beta}{1 - \beta} B. \quad (\text{S43})$$

588 Substituting this into the expected payoff gives a strictly concave function of β ,
 589 with the unique optimum

$$\beta_0^* = \frac{\sum_{j=1}^M \alpha_j^2 \sigma_j^2}{\sum_{j=1}^M \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S44})$$

590 Therefore,

$$c_0^* = \alpha_0 - \frac{B}{\tau^2} \sum_{j=1}^M \alpha_j^2 \sigma_j^2, \quad (\text{S45})$$

591 and

$$L_0 = \frac{\tau^2 \sum_{j=1}^M \alpha_j^2 \sigma_j^2}{\sum_{j=1}^M \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S46})$$

592 We denote the optimal strategy, or the payoff-equivalent strategy class when
 593 $\beta_e^* = 1$, by S_e . A state containing a suboptimal strategy within an interest can
 594 be stationary if the optimal strategy is absent, but it is unstable against invasion
 595 because the optimal strategy has a strictly higher payoff at the same ρ_e . Therefore,
 596 a stable state can contain only S_e within each represented interest.

597 We next consider the frequencies of the $M + 1$ optimal strategies. Let ρ_e denote
 598 the frequency of S_e . Its expected payoff is

$$u_e = -\rho_e L_e. \quad (\text{S47})$$

599 Assuming $L_e > 0$ for all interests, the equal-payoff condition of the replicator
 600 equation at an internal stationary state gives

$$\rho_0^* L_0 = \rho_1^* L_1 = \cdots = \rho_M^* L_M. \quad (\text{S48})$$

601 Together with $\sum_{e=0}^M \rho_e^* = 1$, this uniquely determines

$$\rho_e^* = \frac{D}{L_e}, \quad D = \left(\sum_{j=0}^M \frac{1}{L_j} \right)^{-1}. \quad (\text{S49})$$

602 This internal stationary state is locally stable. At equilibrium, all represented
 603 interests have the same expected payoff, with $\rho_e^* L_e = D$. If the frequency of one
 604 interest increases slightly above its equilibrium value, its payoff decreases because
 605 the niche-expert payoff is proportional to $-\rho_e L_e$. Conversely, if the frequency of
 606 an interest decreases below its equilibrium value, its payoff increases. The replicator
 607 dynamics therefore generate a restoring force that reduces small deviations in interest
 608 frequencies and drives the population back toward the internal stationary state.

609 If one or more interests are absent, the resulting boundary state can be stationary
 610 under the replicator equation, but it is unstable against invasion. The payoff of a rare
 611 absent interest approaches zero as $\rho_e \rightarrow 0$, whereas the mean payoff of the resident
 612 population is negative when $L_e > 0$, so the rare interest increases when introduced
 613 at sufficiently small frequency.

614 Thus, there exists a unique stable full-support equilibrium in terms of the optimal
 615 prediction strategies and their frequencies. Whether the constrained optimum has
 616 $\beta_e^* = 0$, $0 < \beta_e^* < 1$, or $\beta_e^* = 1$ does not change the frequency argument, because
 617 these cases differ only through the resulting value of L_e . When $\beta_e^* = 1$, the strategy
 618 is selectively neutral with respect to c_e , but its prediction, payoff, and equilibrium
 619 interest frequency remain uniquely determined.

620 This equilibrium also allows a direct comparison with the expected accuracy of
 621 the AI. Using Y^{AI} itself without aggregation as the prediction gives

$$\mathbb{E} [(Y - Y^{\text{AI}})^2] = B^2 + \tau^2. \quad (\text{S50})$$

622 For every interest, complete reliance on AI, $\beta = 1$, is a feasible strategy. There-
 623 fore, because L_e is the minimum individual expected squared prediction error within
 624 interest e ,

$$L_e \leq B^2 + \tau^2. \quad (\text{S51})$$

625 Moreover, for interest 0,

$$L_0 = \frac{\tau^2 \sum_{j=1}^M \alpha_j^2 \sigma_j^2}{\sum_{j=1}^M \alpha_j^2 \sigma_j^2 + \tau^2} < \tau^2 \leq B^2 + \tau^2. \quad (\text{S52})$$

626 Because the stable equilibrium has $\rho_0^* > 0$,

$$\sum_{e=0}^M \rho_e^* L_e < B^2 + \tau^2. \quad (\text{S53})$$

627 The collective prediction is the average of individual revised predictions. By
 628 convexity of the squared error,

$$\mathbb{E} [(Y - \hat{Y}^{re})^2] \leq \sum_{e=0}^M \rho_e^* L_e. \quad (\text{S54})$$

629 Therefore,

$$\mathbb{E}[(Y - \hat{Y}^{re})^2] < \mathbb{E}[(Y - Y^{AI})^2]. \quad (\text{S55})$$

630 Because the collective accuracy and AI accuracy use the same normalization by
 631 $\mathbb{E}[(Y - \alpha_0)^2]$, the stationary collective accuracy under the niche-expert structure is
 632 strictly greater than the expected accuracy of the AI prediction.

633 Although the niche-expert structure improves collective prediction relative to the
 634 AI, the stationary collective accuracy does not reach one when $\tau > 0$. For a nonzero
 635 factor i , the residual coefficient of the stationary collective prediction can be written
 636 as

$$\delta_i^* = \alpha_i \{1 - \bar{\beta}^* - \rho_i^*(1 - \beta_i^*)\} = \alpha_i \sum_{e \neq i} \rho_e^*(1 - \beta_e^*). \quad (\text{S56})$$

637 At the stable equilibrium, $\rho_0^* > 0$. In addition, $\tau > 0$ implies $\beta_0^* < 1$. Therefore,
 638 for every factor with $\alpha_i \neq 0$,

$$\delta_i^* \neq 0. \quad (\text{S57})$$

639 Because $\sum_{i=1}^M \alpha_i^2 \sigma_i^2 > 0$, at least one factor contributes positive prediction error.
 640 Hence,

$$\mathbb{E}[(Y - \hat{Y}^{re})^2] > 0, \quad (\text{S58})$$

641 and consequently the stationary collective accuracy satisfies $R^* < 1$.

642 3.2 Chatbot AI

643 3.2.1 Feedback

644 With Chatbot AI, the effect of stochastic error on the collective prediction depends
 645 on n_i , requiring the additional assumption that $n_i \rightarrow \infty$ for every interest for the
 646 stochastic error to vanish. However, Under this assumption, the loss of interest
 647 diversity cannot occur, and therefore the stochastic breakdown of collective accuracy
 648 observed in finite populations cannot be analytically captured. Instead, by assuming
 649 that each interest is represented by infinitely many players, we can show that the
 650 expected collective prediction error monotonically decreases until perfect accuracy
 651 is achieved. Because finite population dynamics are more relevant to the empirical
 652 populations considered in this study, we retain finite population simulations for the
 653 main result.

654 3.2.2 Niche-expert

655 We first consider strategies having the same interest by optimizing the expected
 656 payoff with respect to c and β . For a nonzero interest e , the expected payoff is
 657 strictly concave with respect to c for $\beta < 1$ and $\sigma_e > 0$, and is maximized at

$$c_e^* = \alpha_e. \quad (\text{S59})$$

658 After substituting $c_e^* = \alpha_e$, the payoff is strictly concave with respect to β . Its
 659 unconstrained stationary point is

$$\tilde{\beta}_e = \frac{2\alpha_0 b_e}{2b_e^2 + 2\tau^2}. \quad (\text{S60})$$

660 Because β is constrained between 0 and 1, the optimal β_e^* is $\tilde{\beta}_e$ when $0 < \tilde{\beta}_e < 1$,
 661 and otherwise lies at the corresponding boundary, $\beta_e^* = 0$ or $\beta_e^* = 1$. Because we
 662 assumes AI model with moderate accuracy, b_e and τ is relatively smaller than α_0
 663 which makes $\tilde{\beta}_e > 1$ when $\alpha_0 b_e > 0$ and $\tilde{\beta}_e < 0$ when $\alpha_0 b_e < 0$. These boundary
 664 cases lead to extreme reliance on AI or complete exclusion of AI which both hinders
 665 the emergence of CI. Only when the prediction task is centered near zero and AI is
 666 relatively inaccurate ($\alpha_0 b_e < b_e^2 + \tau^2$), and $\alpha_0 b_e > 0$, the internal optimum β_e^* can be
 667 exist.

668 For interest 0, optimization with respect to c gives

$$c_0^*(\beta) = \alpha_0 - \frac{\beta}{1 - \beta} b_0. \quad (\text{S61})$$

669 Substituting this into the expected payoff gives a strictly concave function of β , with
 670 the unique optimum

$$\beta_0^* = 0. \quad (\text{S62})$$

671 **Supplementary Figure**

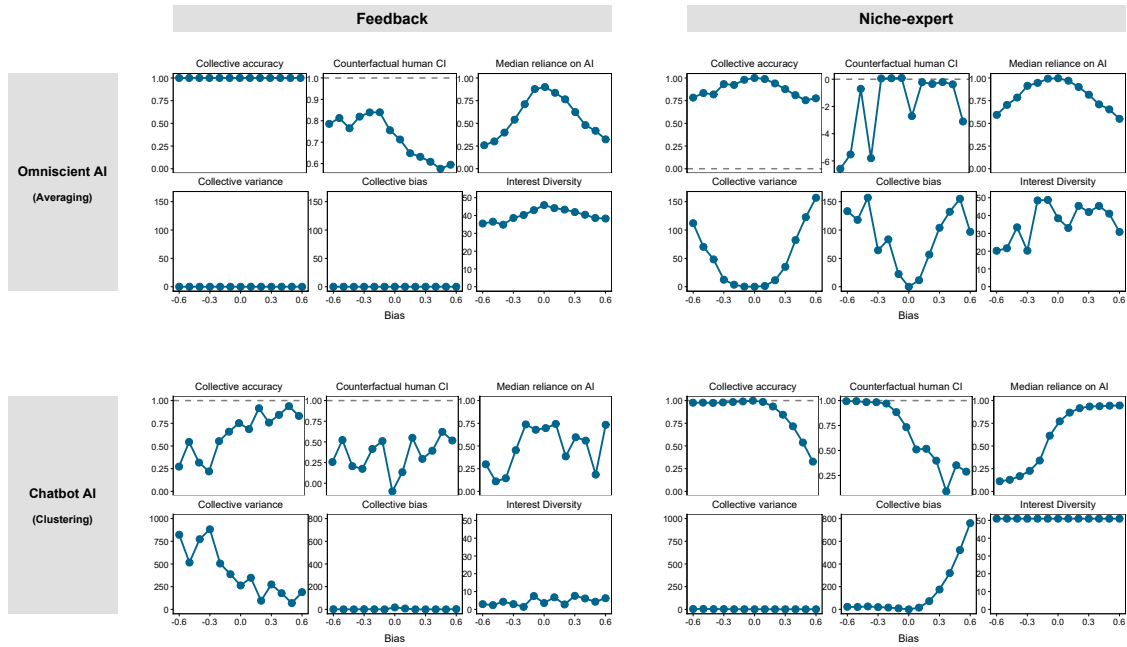


Figure S1: **Caption.** Text.

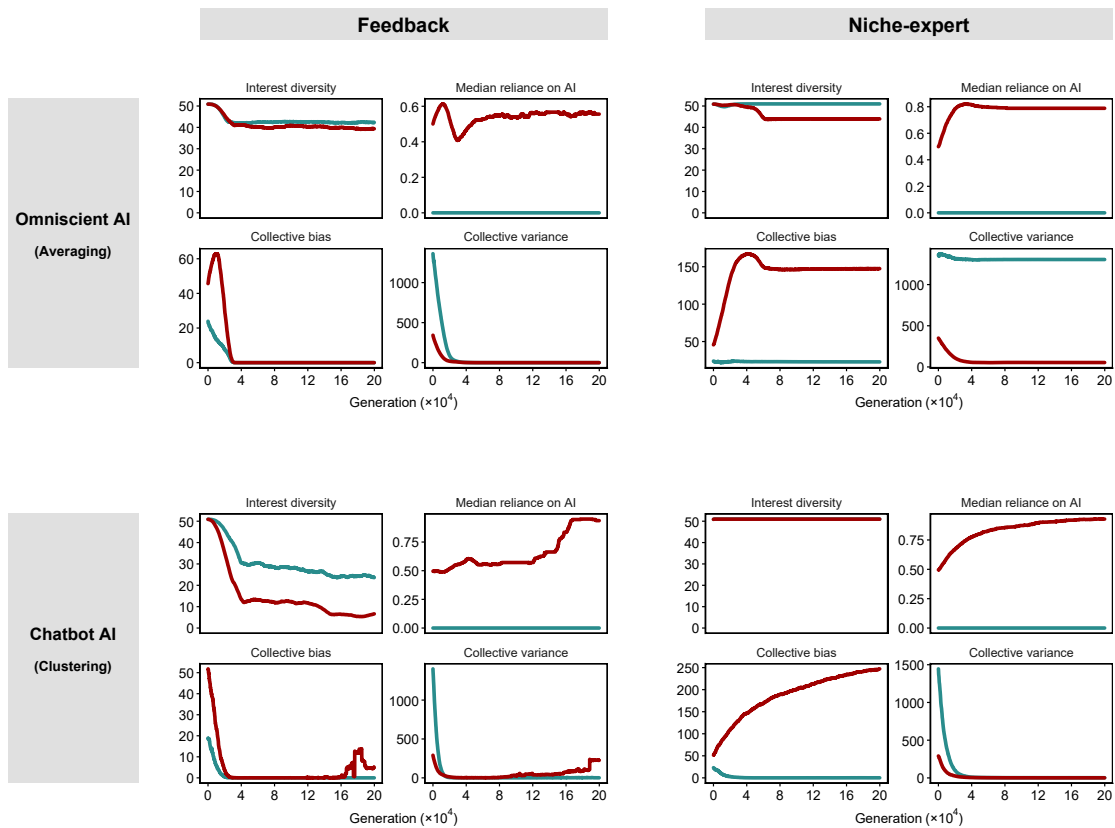


Figure S2: **Caption.** Text.

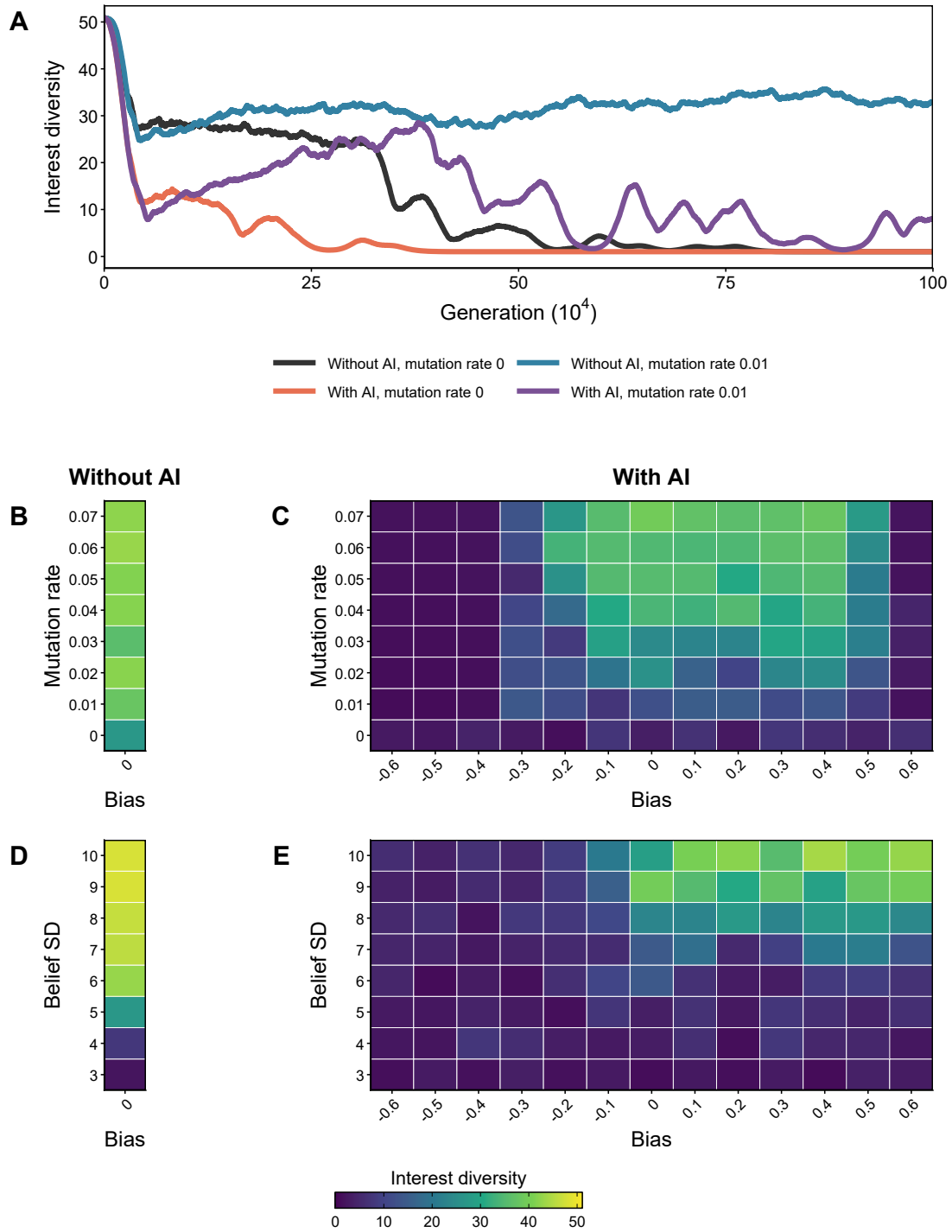


Figure S3: **Caption.** Text.

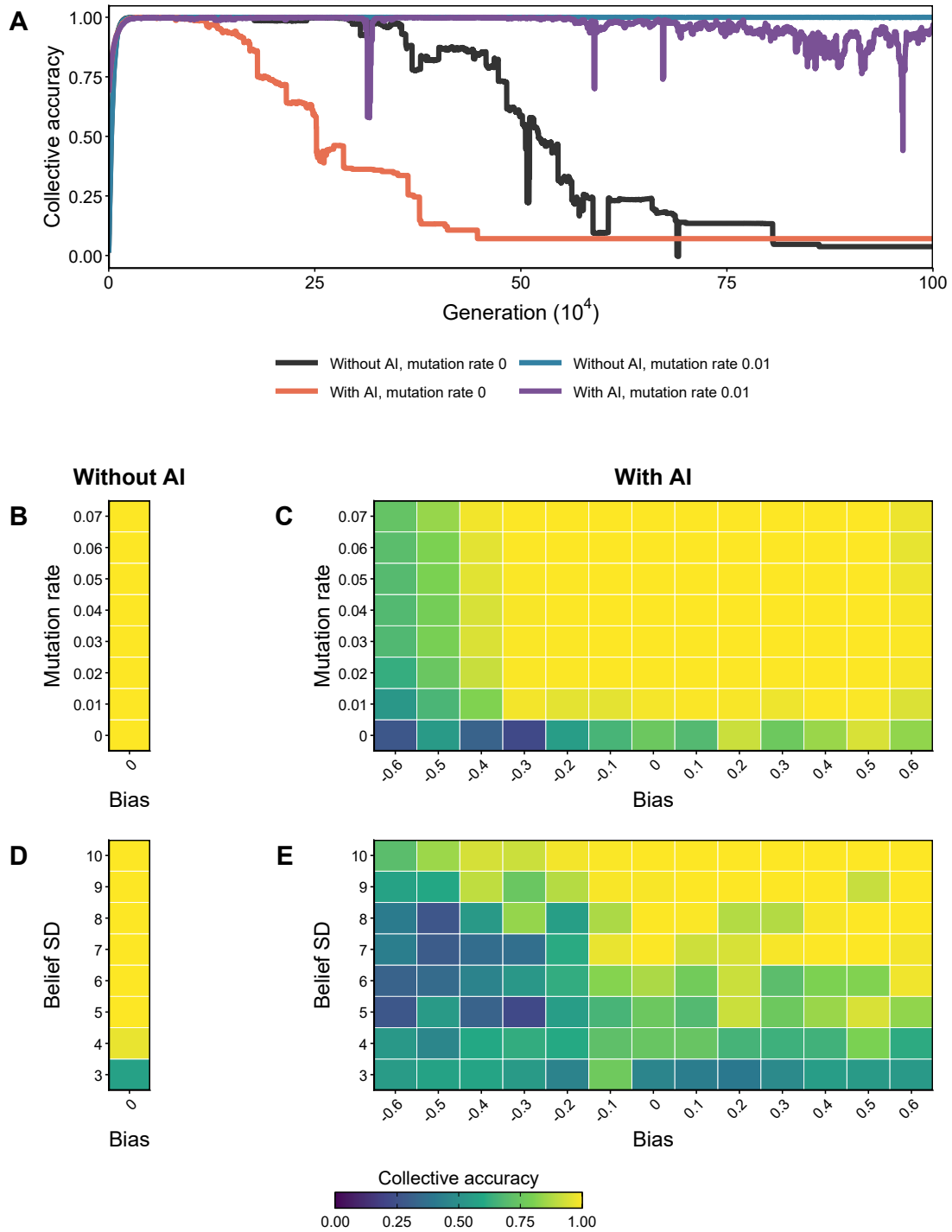


Figure S4: **Caption.** Text.

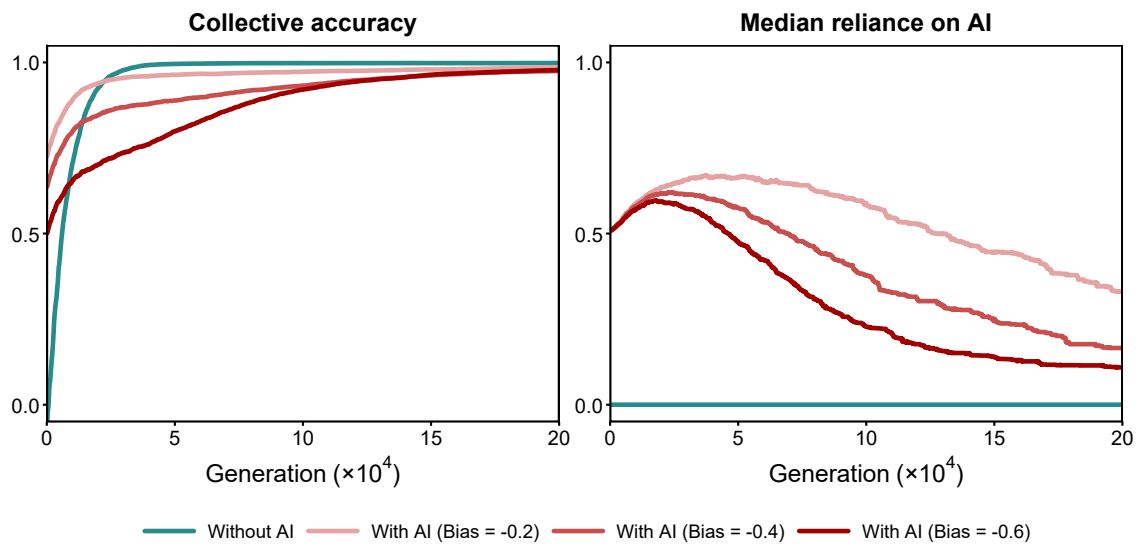


Figure S5: **Caption.** Text.

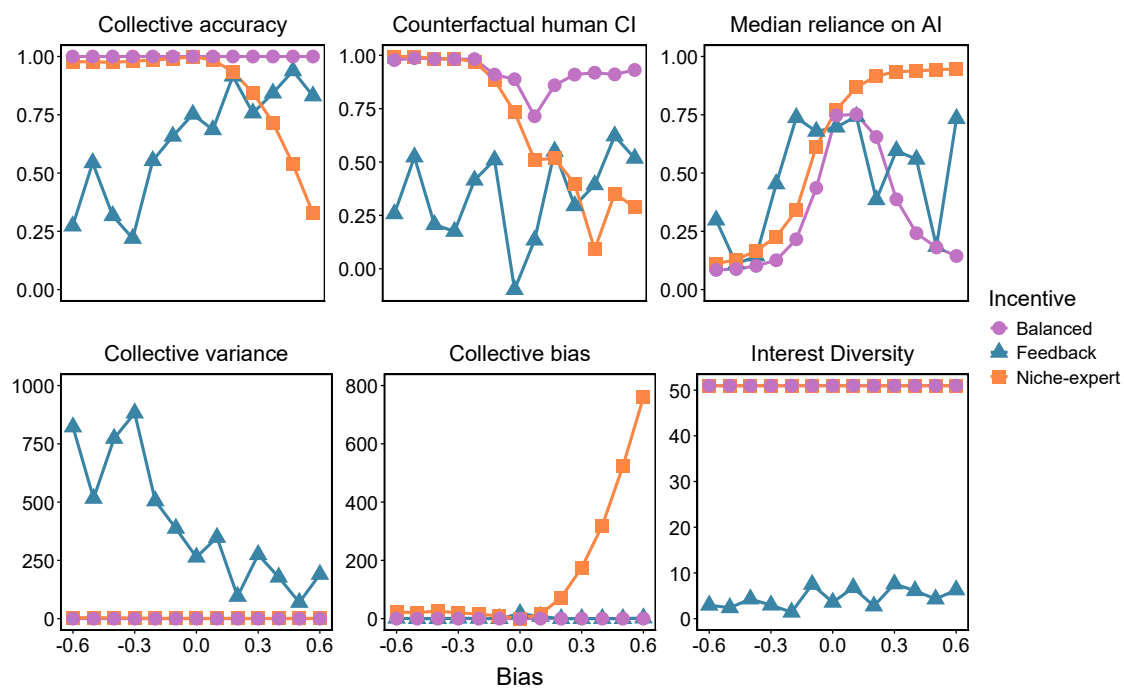


Figure S6: **Caption.** Text.

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